The 10th KAIST-Kyushu University Symposium on Aerospace Engineering
December 6-8, 2018 / KAIST, Daejeon, Korea

Performance Comparison of Neural Networks for Geomagnetic Field Based Indoor Positioning System

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Introduction

For outdoor localization, global navigation satellite system (GNSS) is mainly and widely used. In indoor environments, however, it is difficult to provide precise positioning information for user due to signal loss. Therefore, various method of indoor positioning techniques are being studied. Of the indoor positioning techniques, geomagnetic field based indoor localization using recurrent neural network was studied[1]. We apply several Neural Network techniques to positioning system using geomagnetic field and briefly compared the performance.

Methodology

The geomagnetic field is affected by the steel-frame structure of buildings. This effect causes local variations in geomagnetic field. Through matching these caused local variations and the measured geomagnetic values from current position, the corresponding position can be predicted. To implement this concept, we made the sequential geomagnetic data and then using neural network models to estimate position result.

A. Neural Network model

To analyze which type of model is suitable for geomagnetic based positioning system, training and validation progress was conducted for 3 types of neural networks, basic deep neural network (DNN), gated recurrent unit (GRU) and long short-term memory (LSTM). RNN is an algorithm that reuses state information of the learned neurons using previous data to learn neurons using following data. Basic RNN algorithm works well to train sequence data, but it can store only short-term memory. Such as LSTM, GRU algorithm complement this weakness through using gates in the memory cell.

Table 1 Configuration of used neural network models

Category	Value
Loss function	Mean Square Error (MSE)
Optimizer	Adam
Learning rate	0.001
Activation function	ReLU (DNN), Hard-Sigmoid (GRU, LSTM)

B. Defining Input and Output

In this study, 20 sampled data sequence of geomagnetic 3D field value makes one trajectory. This trajectory corresponding to user movement is used as input. The final outputs of the model are the x, y position value of the last point of a sequence of the input data.

Results and Discussion

A. Geomagnetic Field Map Formation

We acquired the geomagnetic field map of the first floor lobby in KAIST Mechanical Engineering Building using smartphone built in geomagnetic sensor. The geomagnetic field map presents the geomagnetic values in the x, y, and z-axis for each point.

B. Trace Generation

For training, it need trajectory data formulated in geomagnetic field map. We used the random waypoint model which is a random model for the movement of mobile users, and how their location, velocity and acceleration change over time.[2] We made 1,000,000 random waypoint data and each of 1 waypoint data is comprised of x, y, and z-axis geomagnetic field value and the x, y-axis coordinates of location.

Table 2 Configuration of random waypoint model

Parameters	Value
Node's speed	[1.14m/s 1.4m/s] (the pedestrian's average speed range)
Data sampling time	0.5second
Area	13.2m×11.4m

C. Neural Network Models Performance

The performance of the each neural network model is compared by the test set loss in the same configuration and input data. According to Table 3, the loss of GRU model is the smallest. It means that the GRU model is suitable for this positioning system. The LSTM model also presents good performance. The performance is affected by hidden node, but the effect of batch sizes is insignificant.

Table 3 Test loss in terms of the number of hidden node and batch size

hidden node	DNN	LSTM	GRU
16	2.352	0.441	0.432
32	1.733	0.266	0.259
64	1.315	0.212	0.197

batch size	DNN	LSTM	GRU
64	1.711	0.259	0.260
128	1.733	0.266	0.259
256	1.665	0.277	0.270

The following Figure 1 illustrates a portion of the estimated position of the test data set and the true position. Red path represent the true position and blue path represent the estimated position using GRU model. The configuration of used GRU model same as Table 2, and the number of hidden node is 64. From that result, we can confirm that GRU model works well for test dataset.

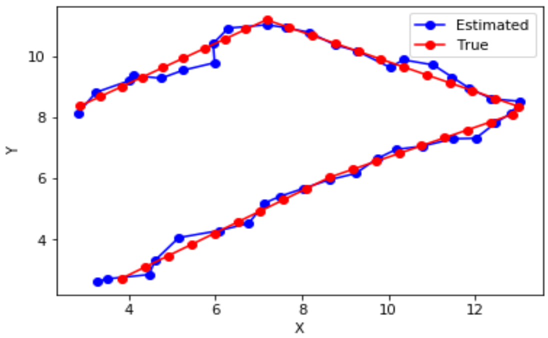

Figure 1 Test results of GRU model

References

[1] Ho Jun Jang, Jae Min Shin and Lynn Choi, "Geomagnetic Field Based Indoor Localization Using Recurrent Neural Networks", IEEE Global Communications Conference, 2017

[2] Mao, Shiwen (2010). "Fundamentals of Communication Networks". *Cognitive Radio Communications and Networks*. pp. 201–234.